ENTERPRISE AI MULTI-AGENT ARCHITECTURE

Enterprise Email Agent Workspace

An autonomous, multi-agent cognitive intelligence platform designed for corporate email triage, concurrent calendar scheduling, automated task persistence, and tone-adapted executive communication.

Capgemini Generative Engine | FastAPI Async Microservice | Microsoft SQL Server (MSSQL) | Microsoft Graph & Outlook | Angular 19 Standalone UI | Native WebSockets (/ws/stream)

5 AUTONOMOUS AGENTS	**5** CORE BUSINESS CASES
0ms SEQUENTIAL LAG (CASE 3)	**100%** HUMAN-IN-THE-LOOP

The Enterprise Challenge vs. The Multi-Agent Solution

Transforming manual inbox friction into structured corporate automation

TRADITIONAL INBOXES (THE PROBLEM)

- ✕ **Cognitive Overload:** Professionals spend 28% of their workday reading, sorting, and contextualizing sprawling email threads.

AGENT WORKSPACE (THE AI SOLUTION)

- ✓ **Automated Ingestion & Noise Stripping:** Reader Agent extracts clean textual context, discarding legal disclaimers and footers.

- ✕ **Manual Calendar Back-and-Forth:** Switching between inbox and Outlook Calendar to cross-reference free/busy slots causes booking delays.
- ✕ **Untracked Action Items:** Critical deadlines buried in informal email bodies get lost without official logging into project boards.
- ✕ **Ambiguity Deadlocks:** Vague emails ("let's connect sometime") waste days in clarifying ping-pong without concrete options.

- ✓ **Live Outlook Availability:** Entity Extractor identifies dates, pulls 2–3 open slots via MS Graph, and stages calendar bookings.
- ✓ **Instant Database Persistence:** Task Extractor parses deliverables, owners, and urgency, saving records directly into MSSQL.
- ✓ **Interactive Clarification & Human Approval:** Supervisor Agent freezes ambiguous flows, generating instant choices for user sign-off.

The Observe → Think → Act Execution Pipeline

Live telemetry broadcast via WebSocket (/ws/stream) with sub-millisecond AgentSpan tracking

1 OBSERVE
Reader Agent
Ingests raw email, strips confidentiality disclaimers, and normalizes payload.

2 THINK
Router Agent
Classifies intent into Meeting, Task, Mixed Intent, or General Inquiry with confidence scores.

3 ACT (CONCURRENT)
Extractors
Queries Outlook free/busy slots and persists to-dos into Microsoft SQL Server.

4 AUDIT
Supervisor Agent
Checks for missing parameters; pauses workflow and creates interactive clarifications if vague.

5 DRAFT & STAGE
Reply Generator
Synthesizes meeting slots and tasks into tone-styled drafts awaiting human approval.

5 Core Enterprise Business Cases

Engineered to handle every nuance of corporate email workflows

CASE 1 — Entity Extractor

The Calendar Finder

Detects scheduling requests (e.g. *"Let's meet tomorrow afternoon"*), queries Outlook Calendar via MS Graph, and presents 2–3 open slots.

- Stage 1: Staged draft for user review
- Stage 2: Booked & confirmed on Outlook Calendar

CASE 2 — Task Extractor

The To-Do List Creator

Scans messages for actionable deliverables (e.g. *"Send the budget deck by Friday 5 PM"*), extracting title, owner, deadline, and urgency.

- Saved directly into MSSQL `Tasks` table
- Urgency: Low | Medium | High | Urgent

CASE 4 — Reply Generator

The Smart Replier

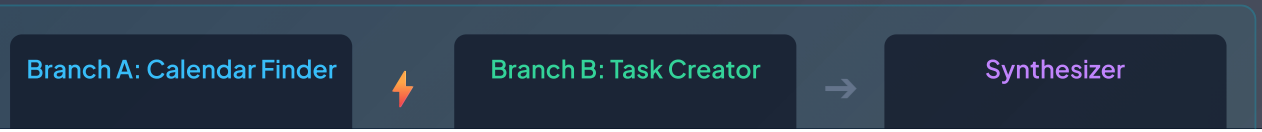
Synthesizes meeting slots and logged task confirmations into a cohesive email reply matching the user's preferred corporate tone.

- Tone: Professional, Friendly, Concise, Executive
- Staged for one-click approval and dispatch

CASE 3 (SPOTLIGHT) — Router & Orchestrator Concurrency

The Multi-Task Router: Mixed-Intent Concurrency

When an email contains **both** a meeting scheduling request and a task assignment (e.g. *"Submit the Q3 budget deck by Friday, and let's meet tomorrow at 2 PM"*), sequential processing causes unacceptable latency. The Orchestrator splits execution branches concurrently using Python's `asyncio.gather`.



CASE 5 — Supervisor Agent

The Double-Checker

Audits completeness. If an email is vague (e.g. *"Let's connect sometime"*), it halts processing, recording a pending clarification with 2–3 selectable choices.

- Workflow paused & flagged in UI
- Resolves upon user answer

Outlook free/busy slots | MSSQL task persistence | Unified Reply Draft

Enterprise Infrastructure & Safety Ecosystem

Robust security boundaries, multi-model fallback, and relational persistence

REST + WS

Capgemini Generative Engine

Multi-model orchestration with automated fallback chain across Nova Micro, Claude Sonnet 5, and GPT-4o.

`amazon.nova-micro` `claude-sonnet-5` `gpt-4o`

ACID Compliant

Microsoft SQL Server

Relational storage for Emails, Tasks, Proposals, Staged Drafts, Telemetry Spans, and Clarification Requests.

`SQLAlchemy 2.0` `ODBC Driver 18` `PyODBC`

Graph API

Microsoft Graph & Outlook

Secure MSAL OAuth2 integration for reading inbox messages, booking calendar events, and mock fallback support.

`MSAL Python` `Outlook Calendar` `Outlook Mail`

Zero Trust

Enterprise Guardrails

Active pre-execution injection pattern blocking, automatic PII masking (credit cards, SSNs), and key scrubbing.

`Prompt Shield` `PII Redaction` `Key Scrubber`

Enterprise Email Agent Workspace v1.0.0 | Capgemini Generative AI Solutions Group